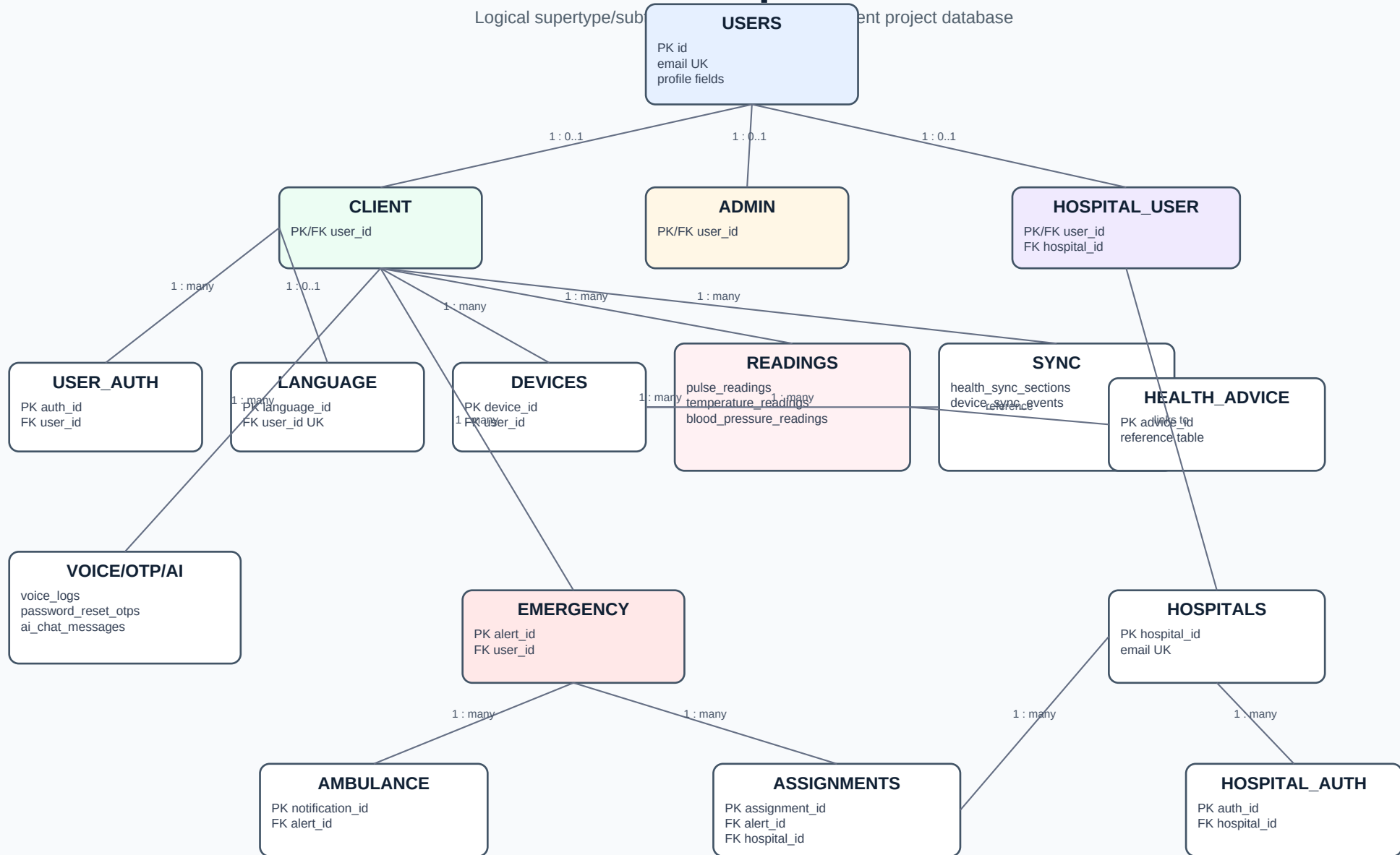


SmartHealth Top-Down EERD

Logical supertype/subtype project database



Note: CLIENT, ADMIN, and HOSPITAL_USER were marked as suggested subtype tables because they were not physical tables in the inspected schema.

SmartHealth EERD Mermaid Diagram

Top-down logical supertype/subtype ERD with users as the root.

erDiagram

```
%% SmartHealth top-down logical EERD.
%% Mermaid ER layout is automatic, but entities are declared from root downward.
```

```
USERS ||--o| CLIENT : "suggested subtype"
USERS ||--o| ADMIN : "suggested subtype"
USERS ||--o| HOSPITAL_USER : "suggested subtype"
```

```
CLIENT ||--o{ USER_AUTH : "has login"
CLIENT ||--o{ LANGUAGE_SETTINGS : "prefers"
CLIENT ||--o{ DEVICES : "owns"
CLIENT ||--o{ PULSE_READINGS : "has"
CLIENT ||--o{ TEMPERATURE_READINGS : "has"
CLIENT ||--o{ BLOOD_PRESSURE_READINGS : "has"
CLIENT ||--o{ HEALTH_SYNC_SECTIONS : "syncs"
CLIENT ||--o{ DEVICE_SYNC_EVENTS : "creates"
CLIENT ||--o{ EMERGENCY_ALERTS : "triggers"
CLIENT ||--o{ VOICE_LOGS : "creates"
CLIENT ||--o{ PASSWORD_RESET_OTPS : "requests"
CLIENT ||--o{ AI_CHAT_MESSAGES : "sends"
```

```
DEVICES ||--o{ PULSE_READINGS : "records"
DEVICES ||--o{ TEMPERATURE_READINGS : "records"
DEVICES ||--o{ BLOOD_PRESSURE_READINGS : "records"
DEVICES ||--o{ HEALTH_SYNC_SECTIONS : "summarizes"
DEVICES ||--o{ DEVICE_SYNC_EVENTS : "logs"
```

```
EMERGENCY_ALERTS ||--o{ AMBULANCE_NOTIFICATIONS : "sends"
EMERGENCY_ALERTS ||--o{ HOSPITAL_ALERT_ASSIGNMENTS : "assigned through"
HOSPITALS ||--o{ HOSPITAL_ALERT_ASSIGNMENTS : "receives"
HOSPITALS ||--o{ HOSPITAL_AUTH : "has login"
```

```
HEALTH_ADVICE }o..o{ PULSE_READINGS : "reference only"
HEALTH_ADVICE }o..o{ TEMPERATURE_READINGS : "reference only"
```

```
USERS {
    int id PK
    varchar email UK
    varchar first_name
    varchar surname
    date dob
    varchar gender
    varchar cell_number
    varchar id_number
    text address
    boolean is_verified
    timestamp created_at
}

CLIENT {
    int user_id PK,FK "suggested, maps to users.id"
}

ADMIN {
    int user_id PK,FK "suggested"
    varchar role "suggested"
}

HOSPITAL_USER {
    int user_id PK,FK "suggested"
    int hospital_id FK "suggested"
}

USER_AUTH {
    int auth_id PK
    int user_id FK
    varchar password_hash
    timestamp created_at
}

LANGUAGE_SETTINGS {
    int language_id PK
    int user_id FK,UK
    varchar language
}

DEVICES {
    int device_id PK
    int user_id FK
    varchar device_type
    varchar manufacturer
    varchar device_model
    varchar platform
    boolean active
}

PULSE_READINGS {
    int pulse_id PK
    int user_id FK
    int device_id FK
    int bpm
    varchar status
    timestamp recorded_at
}

TEMPERATURE_READINGS {
    int temp_id PK
```

```

    int user_id FK
    int device_id FK
    decimal temperature
    varchar status
    timestamp recorded_at
}

BLOOD_PRESSURE_READINGS {
    int bp_id PK
    int user_id FK
    int device_id FK
    int systolic
    int diastolic
    varchar status
    timestamp recorded_at
}

HEALTH_SYNC_SECTIONS {
    int section_id PK
    int user_id FK
    int device_id FK
    timestamp window_start
    timestamp window_end
    decimal heart_rate_average
    decimal temperature_average
}

DEVICE_SYNC_EVENTS {
    int sync_id PK
    int user_id FK
    int device_id FK
    varchar source_platform
    varchar sync_status
}

EMERGENCY_ALERTS {
    int alert_id PK
    int user_id FK
    int bpm
    varchar alert_status
    int countdown_seconds
    timestamp created_at
}

AMBULANCE_NOTIFICATIONS {
    int notification_id PK
    int alert_id FK
    timestamp sent_time
    varchar response_status
}

HOSPITALS {
    int hospital_id PK
    varchar email UK
    varchar name
    varchar phone
    varchar service_area
    boolean active
}

HOSPITAL_AUTH {
    int auth_id PK
    int hospital_id FK
    varchar password_hash
    timestamp created_at
}

HOSPITAL_ALERT_ASSIGNMENTS {
    int assignment_id PK
    int alert_id FK
    int hospital_id FK
    varchar status
    timestamp assigned_at
}

VOICE_LOGS {
    int voice_id PK
    int user_id FK
    text command_text
    timestamp command_time
}

PASSWORD_RESET_OTPS {
    int reset_id PK
    int user_id FK
    varchar otp_hash
    timestamp expires_at
    timestamp used_at
}

AI_CHAT_MESSAGES {
    int chat_id PK
    int user_id FK
    varchar sender_role
    text message
    jsonb vitals_snapshot
    timestamp created_at
}

HEALTH_ADVICE {
    int advice_id PK
    varchar advice_type
    text message
}

```

Plain-English Relationship Explanation

Important finding

The current database does not physically contain client or admin subtype tables. Admin login is handled through environment variables, and patients/clients are stored directly in users. The existing hospital table is hospitals, but it is not currently a subtype of users because it does not have user_id.

Table Classification

Table/entity	Type	Purpose
users	Supertype table	Shared patient/client identity and profile fields.
client	Suggested subtype	Normal app users/patients under users.
admin	Suggested subtype	Staff/admin accounts under users.
hospital_user	Suggested subtype	Connects a users login identity to a hospital account.
user_auth	Transaction/security	Password hashes for users.
language_settings	Lookup/preference	One preferred language per user.
devices	Reference/owned device	Phone/watch/device metadata per user.
pulse_readings	Transaction	Heart-rate readings.
temperature_readings	Transaction	Temperature readings.
blood_pressure_readings	Transaction	Blood pressure readings.
health_sync_sections	Transaction/summary	Mobile health summary windows.
device_sync_events	Transaction/audit	Device sync events.
emergency_alerts	Transaction	Emergency alerts triggered by users.
ambulance_notifications	Transaction	Ambulance notification attempts.
hospitals	Existing hospital table	Hospital organization details.
hospital_auth	Transaction/security	Hospital login password hashes.
hospital_alert_assignments	Junction table	Links emergency alerts to hospitals.
voice_logs	Transaction/audit	User voice command history.
password_reset_otps	Transaction/security	Password reset OTP hashes.
ai_chat_messages	Transaction	AI chat history linked to users.
health_advice	Lookup/reference	Reusable health advice messages.

Relationship Notes

- users to client, admin, hospital_user: logical one-to-one subtype relationship. These were suggested because the current schema does not physically implement them.
- users to user_auth: one-to-many in the current schema because user_auth.user_id is not unique. The app appears to expect one active auth row per user.
- users to language_settings: one-to-one because language_settings.user_id is unique.
- users to devices: one-to-many. One user can own many devices.
- users to readings: one-to-many for pulse_readings, temperature_readings, and blood_pressure_readings.
- devices to readings: one-to-many. A device can record many readings; readings may have a nullable device_id.
- users to emergency_alerts: one-to-many. A client can trigger many emergency alerts.
- emergency_alerts to ambulance_notifications: one-to-many.
- emergency_alerts to hospitals: many-to-many through hospital_alert_assignments.
- hospitals to hospital_auth: one-to-many currently, though logically it should probably be one-to-one.
- users to password_reset_otps: one-to-many.
- users to ai_chat_messages: one-to-many, nullable user link.
- health_advice has no enforced foreign key to readings or alerts; it is a standalone reference table.
- No single health_reading table exists. The system currently models health readings as pulse_readings, temperature_readings, and blood_pressure_readings.